Opened: Friday, 7 November 2025, 11:52 AM
Due: Saturday, 15 November 2025, 11:59 PM

This assignment is divided into two activities:

Activity 2.1 Deduplication:

Attached is a small data set (dedup_data.csv) with about 5K entries each of which is details (given_name, surname, street_number, address_1, address_2, suburb, postcode, state, date_of_birth, soc_sec_id) of synthetically generated persons. Each entry is uniquely identified by the id column and there are potentially multiple entries of the (generated) persons each with some variations in entered details due to reporting noise and errors. The task is to identify all the different records of the same person and group all those records.

Activity 2.2 Crawling:

You are given a web-server you can run locally that you can ping to get a page reporting:

1. A unique identifier for that page (page_id)
2. A node_id that updates at unknown time intervals
3. all the previous node ids and update timestamps for that page, and
4. list of all outgoing links from that page.

Your task is to crawl the website, estimate the page rank of each of the pages and update node ids for all pages efficiently, minimising the number of page visits.

We will be incrementally rolling out updated versions of the web-server and instructions to use the web server.

As a test we have attached a docker image .tar file for amd64 systems you can load and run to launch a web server at port 3000. We have attached a README.md that contains commands for launching the server. **This is not the final version of the web server and may have bugs**. Please reach out to the TAs by replying to the announcement thread on Moodle in case of any issues.

Instructions on running the server

Run the following commands to launch the webserver at http://localhost:3000

Load and Run Image

?

- Verify The Image Contents (One Time Operation. Optional, but recommended)
  - `sha256sum -c crawling_assignment-1.0-amd64.tar.sha256`
- Load the Image (One Time Operation)
  - `docker load -i crawling_assignment-1.0-amd64.tar`
- Run the Image (Every Time you restart the server)
  - `docker run --rm -p 3000:3000 -v $(pwd)/data:/data --tmpfs /tmp:rw,noexec,nosuid --cap-drop ALL --security-opt no-new-privileges --pids-limit 128 --memory 256m crawling_assignment:1.0`

To Stop the Server

- docker stop <container-id-or-name>

Evaluation Endpoint

Overview

The server provides a `/evaluate` endpoint for submitting crawler evaluations. The evaluation window is **60 seconds** from the first visit to the website.

Endpoint

POST `/evaluate`

Request Format

```
{
  "entries": [
    {
      "page_id": "page_abc123",
      "latest_node_id": "xyz789def456",
      "score": 0.0234
    }
  ]
}
```

Fields

- `page_id` (string): The URL path of the page (e.g., "page_abc123")
- `latest_node_id` (string): The most recent node ID you observed for this page
- `score` (number): Your computed PageRank score for this page

Response Format

```
{
  "mse": 0.0012,
  "coverage": 0.85,
  "avg_staleness": 1234.56,
```

```
  "visit_count": 42,
  "matched_entries": 40,
  "total_nodes": 50,
  "covered_nodes": 42
}
```

Response Fields

- `mse` (number): Mean Squared Error between your scores and actual PageRank scores
- `coverage` (number): Fraction of total nodes you covered (0.0 to 1.0)
- `avg_staleness` (number): Average staleness of node IDs in milliseconds
- `visit_count` (number): Total number of page visits at time of evaluation
- `matched_entries` (number): Number of pages successfully matched
- `total_nodes` (number): Total number of nodes in the graph
- `covered_nodes` (number): Number of unique nodes you covered

Requirements

1. **First Visit Required**: You must visit the website at least once before submitting an evaluation
2. **Evaluation Frequency**: Evaluations must be submitted at least every **15 seconds**
   - First evaluation must be within 15 seconds of first visit
   - Subsequent evaluations must be within 15 seconds of the previous evaluation
3. **Evaluation Window**: All evaluations must be submitted within **60 seconds** of the first visit

Error Responses

No Visits

```
{
  "error": "No visits recorded. Please visit the website before submitting an evaluation.
}
```

Status: `400 Bad Request`

First Evaluation Too Late

```
{
  "error": "First evaluation too late. 20.34 seconds have elapsed since first visit. Must
}
```

Status: `400 Bad Request`

Evaluation Gap Too Large

```
{
  "error": "Evaluation gap too large. Last evaluation was 18.45 seconds ago. Must evaluat
}
```

Status: `400 Bad Request`

Metrics Explanation

Mean Squared Error (MSE)

Measures the accuracy of your PageRank scores:

```
MSE = Σ(your_score - actual_score)² / matched_entries
```

Lower is better. A score of 0 means perfect accuracy.

Coverage

Percentage of nodes in the graph that you discovered:

```
Coverage = covered_nodes / total_nodes
```

Range: 0.0 to 1.0 (higher is better)

Staleness

Average age of the node IDs you submitted, measured in milliseconds. Node IDs change periodically, so fresher IDs have lower staleness. **The goal is to minimise staleness and the number of visits.**

Stored Evaluation Data

All evaluations submitted within the 60-second window are stored and encrypted. Each stored evaluation includes:

- `seconds_into_window` (number): Time elapsed since first visit (in seconds with millisecond precision)
- `percentage_of_window` (number): Percentage through the 60-second window (0-100)
- `visit_count` (number): Total visits at time of evaluation
- `mse`, `coverage`, `avg_staleness`: Metric values
- `entries`: Full list of submitted entries

The evaluation data is written to `/data/evaluation.bin` (when running in Docker with the volume mounted) or `evaluation.bin` (when running locally).

	crawling_assignment-1.0-amd64.tar	8 November 2025, 2:43 PM
	crawling_assignment-1.0-amd64.tar.sha256	8 November 2025, 2:43 PM
	dedup_data.csv	16 October 2025, 6:45 PM

Edit submission

Remove submission

Submission status

Attempt number	This is attempt 1.

Submission status	Submitted for grading
Grading status	Not graded
Time remaining	Assignment was submitted 8 days 1 hour early
Last modified	Friday, 7 November 2025, 10:27 PM
File submissions	2024202006_Assignment_2_IRE.rar 7 November 2025, 10:27 PM
Submission comments	Comments (0)